

PathoLens

AI-assisted medical image screening for Malaria, Pneumonia, Breast Cancer and Skin Cancer

Formerly known as the Health Discernment System (HDS)

Overview

PathoLens is a Windows desktop application that analyzes a medical image and screens it for one of four diseases using custom Convolutional Neural Networks (CNNs) built with transfer learning in TensorFlow/Keras. The user uploads an image, the system preprocesses it, runs it through the matching model, and shows the result in plain, user-friendly language.

 **Disclaimer:** PathoLens is an educational / research project. It is a screening aid, **not** a diagnostic tool, and must not replace the judgment of a qualified medical professional.

Supported Screenings and Model Accuracy

Disease	Image Type	Accuracy
Malaria	Blood smear cell images	95.21%
Pneumonia	Chest X-rays	90.47%
Breast Cancer	Histopathology images	86.82%
Skin Cancer	Dermoscopy images	84.77%

Accuracy figures are those reported by the original project team on their test data.

How It Works

1. **Upload:** choose a medical image (`.jpg`, `.jpeg`, `.png`).
2. **Read:** the image is loaded and validated.
3. **Transform:** it is resized and augmented to match the model's input format.
4. **Predict:** the saved CNN builds a feature map and produces a prediction.
5. **Interpret:** the raw output is converted into a clear positive / normal message.
6. **Display:** the result is shown in the interface.

Tech Stack

- **Interface:** Python, Tkinter
- **Models:** Deep Learning, CNNs, Transfer Learning (TensorFlow / Keras)
- **Imaging:** Pillow, NumPy

- **Platform:** Windows (Python 3.6 / 3.7)

Getting Started

1. Run `requirements.exe` once to check and install the required libraries.
2. Run `HDS.exe` to launch the application.
3. Pick a disease module, upload an image (samples are in `test_sample/`), and view the result.

A walkthrough video is included: `HDS_test.mp4`.

■ **Note:** The trained `.h5` model files are not included in this repository.

Project Structure

```
├─ application_ml / _pn / _bc / _sc    # Malaria, Pneumonia, Breast Cancer, Skin Cancer
modules
├─ images/                            # UI assets and screenshots
├─ test_sample/                       # Sample images for each disease
├─ docs/                              # Project report and presentation
├─ HDS.exe, requirements.exe          # Launcher and dependency installer
└─ HDS_test.mp4                       # Demo video
```

Developers

Maintainer (rebrand and documentation):

- **Abuhuraira**

Original creators (Health Discernment System, KIIT Deemed to be University, 2020):

- [Rituraj Saha](#)
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Project guide: Dr. Aleena Swetapadma, School of Computer Engineering, KIIT.

Acknowledgements

Datasets and learning resources: [Kaggle](#), [TensorFlow](#), [SuperDataScience](#), [Udemy](#).